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$\|f-g\|_2 = \left(\int_0^\pi (2\sin^2(x))\,dx\right)^{1/2}$

Now, solve the integral:

$\|f-g\|_2 = \left(\int_0^\pi \sin^2(x)\,dx\right)^{1/2}$

$\|f-g\|_2 = \sqrt{\frac{\pi}{2}}$

So, the distance between $f(x) = \sin(x)$ and $g(x) = -\sin(x)$ on the space $C[0, \pi]$ using the $L^2$ norm is $\sqrt{\frac{\pi}{2}}$.

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# Solution :2

$$\|f-g\|_2 = \left(\int_0^{\pi} |f(x)-g(x)|^2\,dx\right)^{1/2}$$

In this case:

$$\|f-g\|_2 = \left(\int_0^{\pi} |\sin(x)-(-\sin(x))|^2\,dx\right)^{1/2}$$

$$\|f-g\|_2 = \left(\int_0^{\pi} (2\sin^2(x))\,dx\right)^{1/2}$$

Now, solve the integral:

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$$0 \leq x^2 - 2xy + y^2 - 2yz + z^2$$

Rearranging:

$$0 \leq (x-y)^2 + (y-z)^2$$

Since both $d(x,y)$ and $d(y,z)$ are non-negative, this inequality holds.

Therefore, $d(x,y) = (x-y)^2$ satisfies all the properties required to be a metric on the set of real numbers R.

## Solution :2

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4. **Triangle Inequality:** $d(x, z) \leq d(x, y) + d(y, z)$ for all $x, y, z \in R$.

   Let's consider the left-hand side:

   $$d(x, z) = (x - z)^2$$

   And the right-hand side:

   $$d(x, y) + d(y, z) = (x - y)^2 + (y - z)^2$$

   Now, the triangle inequality states that $(x - z)^2 \leq (x - y)^2 + (y - z)^2$.

   Expanding both sides:

   $$(x - z)^2 \leq x^2 - 2xy + y^2 + y^2 - 2yz + z^2$$

   Simplifying:

   $$0 \leq x^2 - 2xy + y^2 - 2yz + z^2$$

   Rearranging:

   $$0 \leq (x - y)^2 + (y - z)^2$$

   Since both $d(x, y)$ and $d(y, z)$ are non-negative, this inequality holds.

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In this case, $(x-y)^2$ is always non-negative since it involves squaring a real number. So, $d(x,y) = (x-y)^2$ satisfies non-negativity.

2. **Identity of Indiscernibles:** $d(x,y) = 0$ if and only if $x = y$ for all $x, y \in R$.
   If $x = y$, then $(x-y)^2 = 0^2 = 0$. Conversely, if $(x-y)^2 = 0$, then $x - y = 0$, and thus $x = y$
   Therefore, the identity of indiscernibles holds.
3. **Symmetry:** $d(x,y) = d(y,x)$ for all $x, y \in R$.
   In this case, $(x-y)^2 = (y-x)^2$ since squaring eliminates the sign. Thus, symmetry is satisfied.
4. **Triangle Inequality:** $d(x,z) \leq d(x,y) + d(y,z)$ for all $x, y, z \in R$.
   Let's consider the left-hand side:
   $d(x,z) = (x-z)^2$
   And the right-hand side:
   $d(x,y) + d(y,z) = (x-y)^2 + (y-z)^2$